

B03 Practical Mastery Check

Closed-book dimensional-model transfer - non-gating

Non-gating closed-book mastery check

This check does not unlock, block or replace Gate A. Save a genuine attempt before opening the protected review.

Scenario - Parcel Delivery

- Using only Practice_Data, classify customers, services, shipments, shipment_events, monthly_targets and customer_tags as dimensions, facts or bridge/factless-fact candidates. State one-row grain and key for each.
- Draw or describe the relationship paths. State one-side uniqueness, many-side FK and intended filter path. Explain why customer_tags is a legitimate membership bridge.
- Explain why shipments and shipment_events must remain separate facts and predict what happens to booked_revenue if events are raw-joined before aggregation.
- Declare booking_date, pickup_date and delivery_date roles and how a shared Date dimension would be reused without pretending the three meanings are interchangeable.
- Explain why monthly_targets is a higher-grain fact and how you would compare target vs shipment actuals at a compatible month-region grain.
- Write a reconciliation checklist: source rows, distinct transaction keys, event coverage, revenue total, target rows, dimension uniqueness/FK checks and fan-out tests.
- Produce a short implementation handoff and one rejected alternative design with the reason it was rejected.

Evidence to save before review

- Process/grain/key/role table.
- Relationship/bridge/date-role sketch.
- Raw controls and a fan-out test.
- Reconciliation checklist with pass/fail evidence.
- Rejected alternative with reason.
- Own-words explanation of what Course 3 will implement later.

Assessment rule

This is mastery evidence only when the model is internally consistent and the raw controls reconcile. A polished diagram with mixed grain or unexplained fan-out is not a pass.